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N. R. Comins

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The Optical Properties of Liquid Metals

By N. R. COMINS

Cavendish Laboratory, Free School Lane,
Cambridge, U.K.†

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ABSTRACT

The optical constants of the liquid metals Cu, Au, Hg, Ga, Al, Sn, Pb, Bi and of the Cu-Sn binary alloy system have been measured by an ellipsometric method at wavelengths, mainly in the infra-red region, of up to $8\text{ }\mu\text{m}$ and at temperatures up to 1200°C . The results are compared with those obtained by previous workers at shorter wavelengths, mainly in the visible. Except in the case of Hg they are in satisfactory agreement with theoretical predictions based upon the nearly-free-electron model.

§ 1. INTRODUCTION

THE optical properties of liquid metals have been extensively investigated already in the spectral range from the near infra-red to the near ultra-violet, i.e. at wavelengths between about $3\text{ }\mu\text{m}$ and $0.3\text{ }\mu\text{m}$ (Schulz 1957, Hodgson 1959, 1960, 1961, 1962, Smith 1967, Miller 1969). The results have usually been expressed in terms of the real and imaginary parts of a complex dielectric constant $\hat{\epsilon}$ ($=\epsilon_1 + i\epsilon_2$) and have been fitted quite successfully by the Drude formulae

$$\epsilon_1 = 1 - \frac{n^*e^2\tau}{m\epsilon_0(1 + \omega^2\tau^2)}, \quad \epsilon_2 = \frac{n^*e^2\tau}{m\epsilon_0\omega(1 + \omega^2\tau^2)} \quad \dots \quad (1)$$

after some adjustment of the effective carrier density n^* and the relaxation time τ . The closeness of the fit achieved, and the absence of the sort of structure in the curves for ϵ_1 and ϵ_2 that is associated with interband absorption in solid metals, have provided much support for the nearly-free-electron model of liquid metals, upon which Ziman's theory (1961) of their electrical transport properties is based.

This paper describes optical measurements at wavelengths of up to $8\text{ }\mu\text{m}$ and at temperatures of up to 1200°C . The motivation for the work was three-fold. First, it seemed desirable to investigate some systematic deviations from the Drude curves which were detectable in many of the published results at wavelengths near the limit of measurement in the near infra-red (e.g. in Hodgson's figures for Sn, Ge, and Bi). Secondly, it

† Present address : National Research Council of Canada, Ottawa, Canada.

was noted that the fitted curves for $\sigma(\omega)$ ($= \epsilon_0 \omega \epsilon_2$) appeared to extrapolate, in nearly every case, to a value $\sigma(0)$ ($= n^* e^2 \tau / m$) which was distinctly less than the bulk d.c. conductivity $\sigma_{d.c.}$. It was hoped to study the way in which the curves for $\sigma(\omega)$ climb up at low frequencies, as they surely must, towards $\sigma_{d.c.}$, and hence to gain an insight into the origin of this anomaly. Thirdly, it was of interest to study Hg in the infra-red, since for this particular metal ellipsometric measurements at shorter wavelengths have yielded results which are not Drude-like at all.

In the event the results proved somewhat negative in character: no significant deviations were observed in the neighbourhood of $3\mu\text{m}$ and those reported by previous workers are now thought to have been spurious; the curves for $\sigma(\omega)$ turned out to remain remarkably flat down to the lowest frequencies accessible; and the behaviour observed for Hg, though distinctly anomalous, was not quite so dramatic as an isolated measurement by Beattie and Conn (1955) at a wavelength of about $6\mu\text{m}$ had prepared us to expect. Perhaps the most useful outcome, therefore, is a set of values for n^*/n more accurate than those hitherto available. The significance of these values is briefly discussed in § 4.

§ 2. EXPERIMENTS

2.1. *Principles of the Method*

The measurements were made using a variation of Beattie and Conn's ellipsometric method (Beattie and Conn 1955) developed by Miller (1969). A collimated beam of light was passed through a polarizer, reflected from the free surface of the liquid metal specimen at a large angle of incidence, and the resultant elliptically polarized light passed through an analyser, monochromator and then onto the detector. The basis of the method is to find the analyser settings Q for which equal intensities are obtained with fixed polarizer settings of 0 and $\pi/4$ or alternatively with $\pi/4$ and $\pi/2$. Ideally, these angles satisfy the equation,

$$\tan^2 Q - 2\rho \cos \Delta \tan Q - \rho^2 = 0, \quad . \quad . \quad . \quad . \quad . \quad (2)$$

where $\rho \exp(i\Delta)$ is the ratio between the amplitude reflection coefficients for the p and s-components. It is straightforward to obtain ρ and Δ from the two different angles Q_1 and Q_2 with the above property, since

$$\begin{aligned} \rho^2 &= \tan |Q_1| \tan |Q_2| \\ \cos \Delta &= \frac{1}{2\rho} [\tan |Q_1| - \tan |Q_2|]. \quad . \quad . \quad . \quad . \quad (3) \end{aligned}$$

The values of ϵ_1 and ϵ_2 may be obtained from ρ and Δ using the standard Fresnel equations (e.g. Faber and Smith 1968). The method has the advantage that the detector does not have to be linear, nor need it be insensitive to polarization. If the light from the source is slightly polarized,

as it was in the present work, it is a straightforward matter to apply corrections to eqn. (3). Slow zero drifts have no effect on the results and this is particularly useful since vibrations of the specimen surface necessitate the use of fairly long time-constants.

It is well known that the optical constants are very sensitive to small variations in ρ and Δ and great care must therefore be exercised to eliminate errors. To eliminate (to first order) the effects of errors in setting the zeroes of the polarizer and analyser scales, all the measurements were repeated with polarizer settings of 0 and $-\pi/4$ and also $-\pi/4$ and $-\pi/2$, as well as those mentioned earlier, and the results for corresponding pairs averaged. To reduce discrepancies caused by irregularities in the polarizing films, care was taken to centre the beam and to check that the intensity was unchanged by rotations of π . In some spectral regions corrections were applied for incomplete polarization by the polarizers.

The accuracy of each measurement of Q_1 and Q_2 was usually limited to about 0.1° by fluctuations of intensity caused by vibration of the specimen surface, and, to within 0.1° repeated measurements were usually in agreement. The accuracy of the final results for ϵ_1 and ϵ_2 depended considerably on the metal and on the wavelength. In general the sensitivity to variation in Q_1 and Q_2 increased with increasing wavelength, but in the case of the 4 and 5-valent metals, ϵ_1 was particularly sensitive. Additional errors of a systematic nature may arise if the beam is slightly convergent so that not all of it is incident upon the surface at the mean angle $\bar{\theta}$ which one is obliged to use in solving the Fresnel equations (see Hodgson 1955). In the present work, these additional errors should not exceed 3% for both ϵ_1 and ϵ_2 .

2.2. Experimental Details

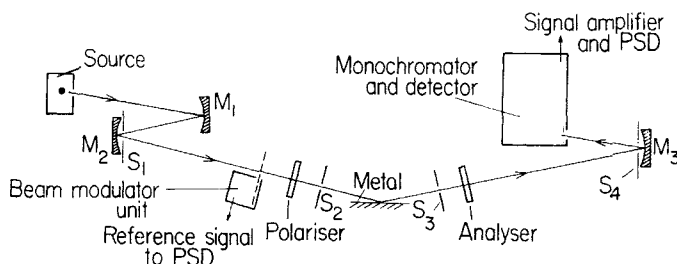
(a) The optical system

The basic optical system is shown in fig. 1. The beam was carefully defined by the four slits, each arm of the ellipsometer being 1 m long. To determine the angle of incidence accurately, it was necessary to measure the angles made by both incident and reflected beams with the horizontal. These angles were not always equal, depending on the curvature of the liquid metal surface. The angle of incidence in the infra-red region was set at about 84° with a beam spread of 0.15° about the central ray. For work at shorter wavelengths the angle was reduced. The mirror system M_1 and M_2 provided a 1 : 1 image of the source on the sample and M_3 focused the reflected light onto the entrance slit of the monochromator.

The source used for wavelengths greater than $3\mu\text{m}$ was a Nernst glower enclosed in a water-cooled jacket to prevent fluctuations due to air currents. For wavelengths less than $3\mu\text{m}$, it was preferable to use the brighter tungsten-iodide lamps.

Two types of polarizers were used in the experiments. In the range 0.9 to $2\mu\text{m}$ HR-Polaroid films were used, while from 3.5 to $8\mu\text{m}$ these were replaced by Perkin-Elmer wire-grid polarizers. The monochromator

Fig. 1



The ellipsometer. M_1 , M_2 and M_3 are front-aluminized spherical mirrors. S_1 , S_2 , S_3 and S_4 are slits to define the beam.

used in the range 3 to $8\ \mu\text{m}$ contained a CaF_2 prism in the Littrow mounting. The resolution was adjusted to be about $0.1\ \mu\text{m}$. For wavelengths less than $3.5\ \mu\text{m}$, a series of narrow-band transmission filters were used instead.

The detector was a Pye Unicam SP50 Golay system. The incident beam was mechanically chopped at a frequency of $11\ \text{Hz}$, and the Beam Modulator Unit also provided a synchronous reference signal for the phase-sensitive detector. By using chopped light the effects of emission from the specimen at high temperatures were eliminated. For some experiments a PbS detector was used in the 1 to $2\ \mu\text{m}$ range.

(b) *The cells*

Two cells were used in the experiments. The one used for high temperature work consisted of a water-cooled stainless steel can with removable ends sealed by 'O'-rings. Inside the can was a heating coil surrounded by radiation shields and inside that a flat dish to hold the specimen. Alumina dishes were suitable for most of the metals investigated, but Mo dishes were preferred for the work on Cu, and a W one was used with Ga and for one experiment on Sn. It is important to choose a dish in which the meniscus of the specimen is reasonably flat, since otherwise, the reflected intensity is much reduced. The temperature of the specimen was measured by two chromel-alumel thermocouples.

For the experiments on Hg a glass cell, similar in design to that used by Faber and Smith (1968), was used. The cell consisted basically of a glass bulb with a glass dish incorporated in it and with two arms carefully positioned to allow the entry and exit of the light beam. In the near infra-red region it was possible to use Pyrex windows fused onto the arms, but at larger wavelengths it became necessary to use CaF_2 windows and 'O'-rings. The Hg samples were distilled directly into the dish under vacuum.

Both cells were connected to a standard vacuum system capable of producing pressures lower than 10^{-5} torr and were mounted on rubber anti-vibration supports in an attempt to minimize vibrations of the

specimen surface during pumping. Care was taken to ensure that the light beam was normal to the windows to prevent spurious polarization effects. No effects of stress birefringence were observed even when the windows were under vacuum.

(c) *Specimen preparation*

The pure metals investigated were Cu, Au, Al, Ga, Sn, Pb, Bi and Hg. Except in the case of Hg, they were introduced into the apparatus as solid ingots, after a thorough etch and wash to remove surface contamination. Wherever possible, they were melted under vacuum and allowed to outgas at a temperature which depended upon their vapour pressure. After this initial treatment the specimens were often covered with an oxide layer and in this state could not be used for the experiments. Dry hydrogen was then leaked into the system, up to a pressure of between 1 and 300 torr depending on the evaporation rate of the metal, and the temperature slowly increased until the oxide layer disappeared. The hydrogen helped to prevent evaporation as well as to make the surface clean. No difficulties were experienced in obtaining clean reproducible surfaces in this way, except in the case of Al, which had to be heated to about 900°C. It is believed that at this temperature the oxide layer on Al dissolves in the metal (Powell 1968) and at higher temperatures the surface did become clean to the eye.

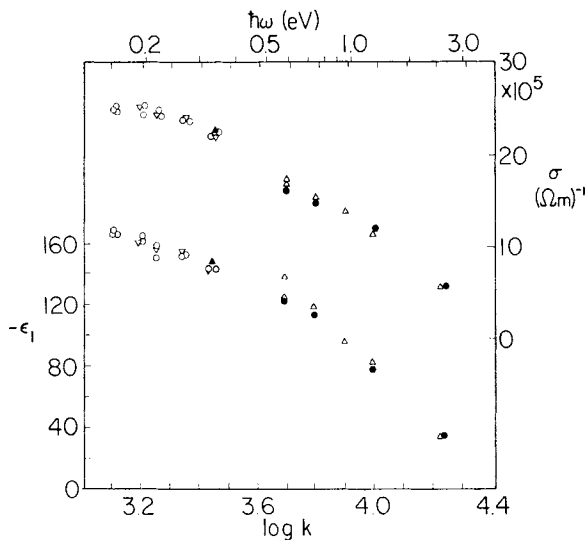
At least three different samples of each pure metal were investigated, except for Au where only one was available. Each experiment lasted for several days, during which the specimen was solidified and remelted a number of times. No systematic changes with time were observed.

Experiments were also carried out on three Cu-Sn alloys. They were prepared *in situ* by melting separate ingots of Cu and Sn in the dish and, after considerable shaking, leaving them overnight to diffuse together. Their concentration was determined by weighing the ingots initially. Fortunately, the vapour pressures of Cu and Sn are almost identical at 1150°C so the concentration did not change with time. Like the pure specimens, these alloys could be frozen and remelted without their optical properties changing.

§ 3. RESULTS

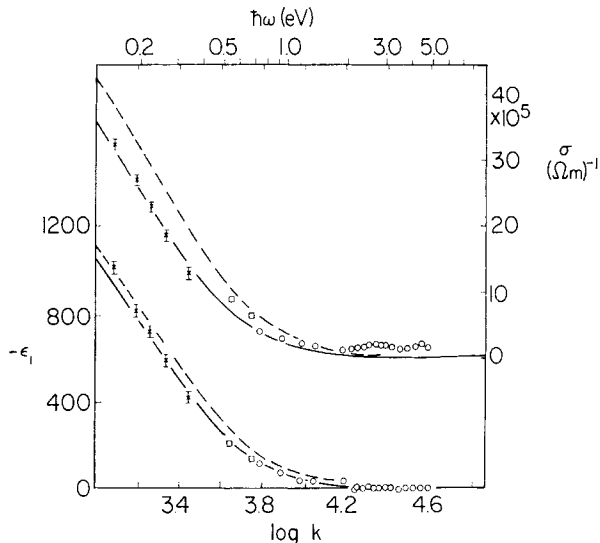
A selection of the results obtained for pure liquid metals are displayed in figs. 2 to 9, where $\epsilon_1(\omega)$ and $\sigma(\omega)$ are plotted against $\log_{10} k$ (k being the inverse of the free-space wavelength, expressed in cm^{-1}) and also against photon energy. Figure 2 contains raw data, to illustrate the extent to which results from separate experiments are liable to disagree; the spread is no greater than was to be expected in view of the sources of error discussed in § 2.1. The later figures show only the averaged results for each wavelength, with error bars to indicate the range covered by the individual readings.

Fig. 2



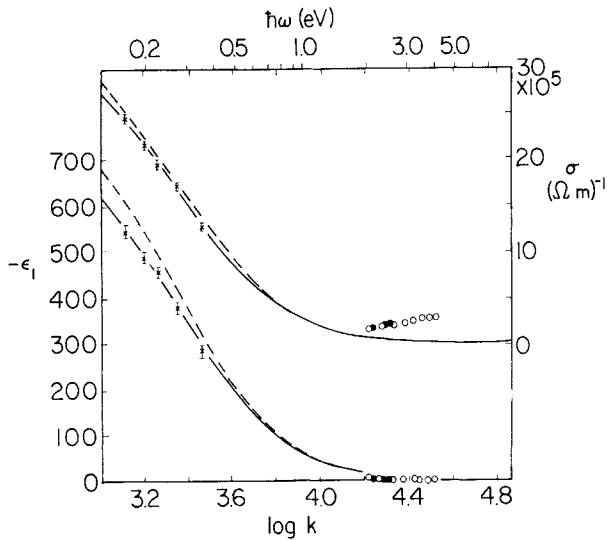
The experimental results for σ and ϵ_1 for liquid gallium (600°C) showing values from five different specimens.

Fig. 3



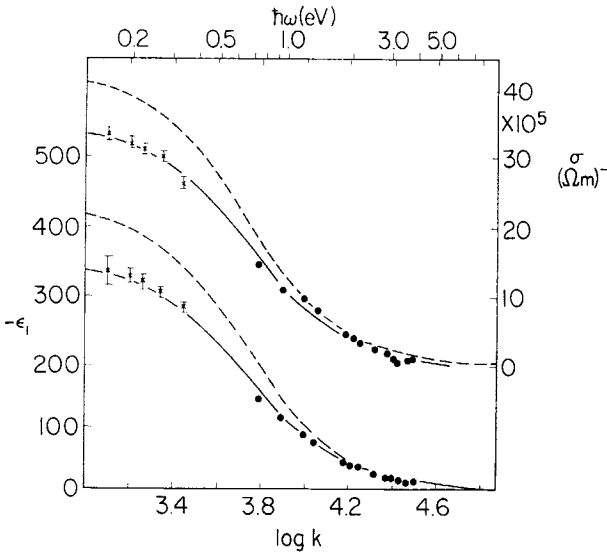
The optical constants of liquid copper (1100°C): $\times$ (present work), $\square$ (Hodgson 1960) and $\circ$ (Miller 1969). The conductivity σ and dielectric constant ϵ_1 are plotted against $\log_{10} k$ (k being the inverse of the free-space wavelength, expressed in cm^{-1}) and also against photon energy. The full lines represent the fitted curves, while the broken curves show the behaviour predicted by the Drude formula with $n^*/n=1$ and $\sigma(0)/\sigma_{\text{d.c.}}=1$.

Fig. 4



The optical constants of liquid gold (1120°C): × (present work), ● (Otter 1961), ○ (Miller 1969).

Fig. 5



The optical constants of liquid aluminium (900°C): × (present work) and ● (Miller 1969).

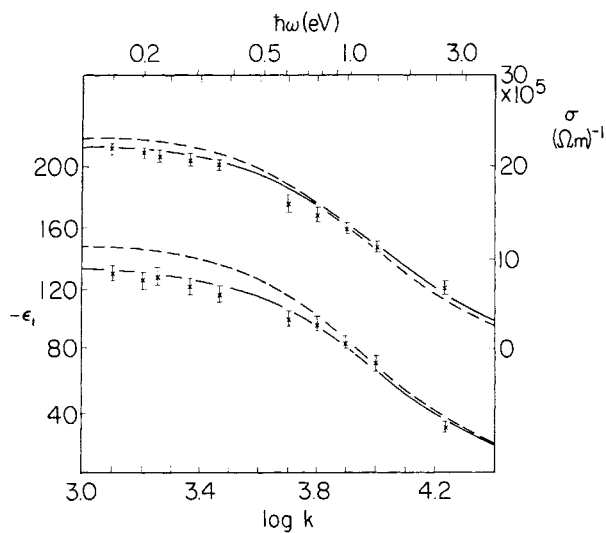
The results are in reasonable agreement with those of previous workers ; Hodgson's two points for Cu, which are included in fig. 3 and which do not conform to the present data, were measured under adverse conditions and should probably be ignored. The points lie on smooth curves and there is no evidence of interband absorption by the valence electrons. The absorption peak which is evident for both Cu and Au beyond 2 eV is well known to be associated with excitation of d electrons from the ion core.

The broken curves in figs. 3 to 8 are computed from the Drude formulae with n^* set equal to the density of valence electrons n , and with τ chosen so as to make $ne^2\tau/m$ equal to the measured d.c. conductivity at the temperature concerned. The full curves correspond to the Drude formulae with n^* and τ adjusted to give the best overall fit. An allowance for core polarizability, computed from Miller's (1969) absorption data in the ultra-violet with the aid of the Kramers-Kronig relation, has been included

Table 1. Parameters used in fitting curves

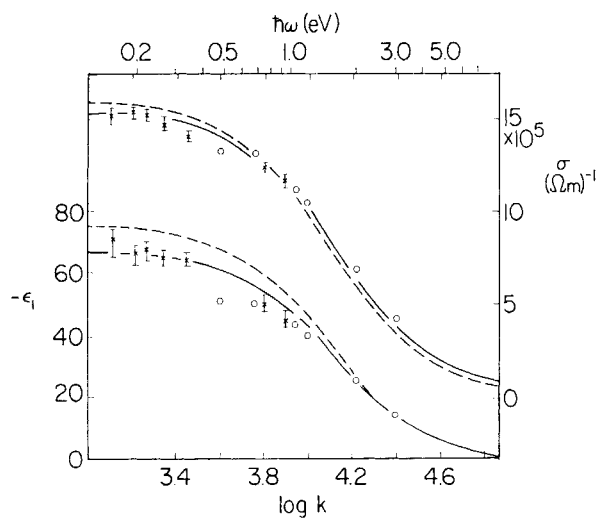
Metal	Valency	Temp (°C)	n^*/n	$\sigma(0)/\sigma_{d.c.}$	Reference
Copper	1	1100	0.76 ± 0.04	0.88 ± 0.02	Present work
Gold	1	1100	1.01 ± 0.02	0.98 ± 0.01	Present work
Aluminium	3	900	0.88	0.75	Miller (1969)
		900	0.83 ± 0.02	0.83 ± 0.02	Present work
Gallium	3	1170	0.80 ± 0.02	0.84 ± 0.02	Present work
		—	0.98	0.98	Schulz (1957)
		600	1.03 ± 0.03	0.96 ± 0.02	Present work
Tin	4	860	1.00 ± 0.03	0.98 ± 0.02	Present work
		400	1.02	1.04	Kent (1919)
		446	1.17	0.94	Hodgson (1961)
		858	1.15	0.93	Hodgson (1961)
		600	1.10 ± 0.02	0.98 ± 0.02	Present work
Lead	4	800	1.05 ± 0.02	0.96 ± 0.02	Present work
		1170	1.03 ± 0.03	0.96 ± 0.02	Present work
		400	1.28	0.96	Kent (1919)
		390	1.25	1.02	Smith (1967)
		567	1.15	0.95	Hodgson (1961)
		786	1.14	0.93	Hogsdon (1961)
Bismuth	5	600	1.19 ± 0.03	0.92 ± 0.02	Present work
		800	1.13 ± 0.04	0.95 ± 0.03	Present work
		400	1.02	0.96	Kent (1919)
		340	1.08	1.01	Smith (1967)
		505	1.06	0.96	Hodgson (1962)
		695	1.10	0.98	Hodgson (1962)
		500	1.05 ± 0.04	0.93 ± 0.03	Present work
Copper-Tin alloys					
30% Cu	3.1	1150	1.02 ± 0.03	0.97 ± 0.02	Present work
70% Cu	1.9	1150	0.91 ± 0.05	0.96 ± 0.02	Present work
90% Cu	1.3	1150	0.98 ± 0.05	0.95 ± 0.03	Present work

Fig. 6



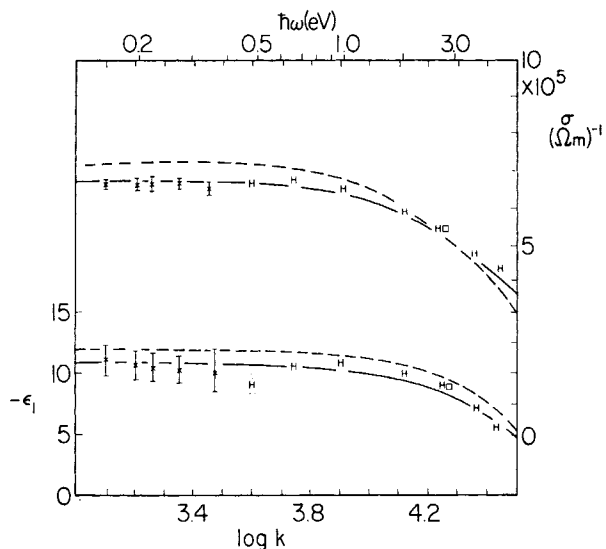
The optical constants of liquid gallium (860°C): × (present work).

Fig. 7



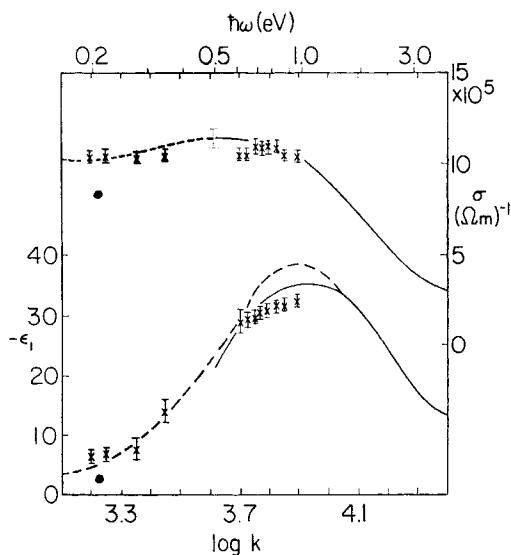
The optical constants of liquid tin (800°C): × (present work) and ○ (Hodgson 1961).

Fig. 8



The optical constants of liquid bismuth (500°C): $\times$ (present work), H (Hodgson 1962) and $\square$ (Kent 1919).

Fig. 9



The optical constants of liquid mercury at room temperature: $\times$ (present work), $\bullet$ (Beattie and Conn 1955) and the solid line indicating the averaged results of Hodgson (1959), Smith (1967), Busch and Guggenheim (1968). The broken line is computed from the semi-empirical formula of Smith (1968).

in the curves for $\epsilon_1(\omega)$ in Cu and Au; for the other metals investigated the effects of core polarizability are small enough to be ignored. Table 1 lists all the fitting parameters used for the full curves shown in the figures and for similar sets of results which there is no space to display, and also by previous experimenters. The values used for the density and for $\sigma_{\text{d.c.}}$ were taken from the Liquid Metals Handbook (1952) or from references quoted by Cusack (1963).

In fig. 9 for Hg, it would be pointless to attempt to fit Drude curves. In this figure full lines are used to indicate the extensive ellipsometric results of Hodgson (1959), Smith (1967) and Busch and Guggenheim (1968), while the broken curves are computed from the semi-empirical formulae of Smith (1968). Since Smith's formulae for σ and ϵ_1 are consistent with the Kramers-Kronig relations, it can be seen by inspection that the experimental results, despite their anomalous character, are also consistent in this sense. The isolated point due to Beattie and Conn (1955), which suggested a surprising minimum in $\sigma(\omega)$ at around 0.2 eV and perhaps an absorption edge between 0.2 and 0.5 eV is clearly erroneous.

The alloy results are summarized both in table 1 and in figs. 10 and 11. The 'effective valency', Z^* , which is shown in fig. 10 to be a linear function of atomic concentration, is defined by

$$Z^* = n^* \frac{V}{N} = \frac{n^*}{n} (c_{\text{Cu}} Z_{\text{Cu}} + c_{\text{Sn}} Z_{\text{Sn}}), \quad . \quad . \quad . \quad . \quad . \quad (4)$$

where N is the total number of atoms in volume V , while Z_{Cu} and Z_{Sn} are the valencies of the two components (assumed to be 1 and 4, respectively) and c_{Cu} and c_{Sn} are their concentrations. Figure 11 shows that $\sigma(0) (= n^* e^2 \tau / m)$ follows the curve obtained for $\sigma_{\text{d.c.}}$ by Roll and Motz (1957), though it is uniformly lower by a few parts per cent.

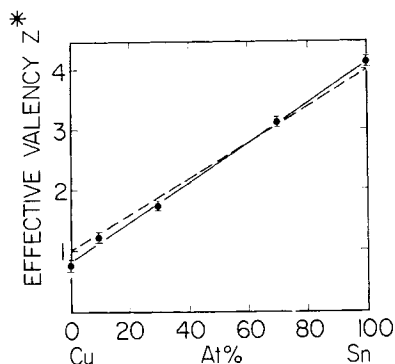
§ 4. DISCUSSION

The theory of the optical properties of liquid metals has been extensively discussed by Faber (1972). He comes to the conclusion that, if there is no residue of interband absorption in the liquid state and if—as recent theoretical work suggests (e.g. Shaw and Smith 1969)—the density of states near the Fermi level is very little affected by those structure-dependent terms in the energy of the conduction electrons which involve the ionic pseudopotential *to second order*, then we may indeed expect the Drude formulae to hold as a good approximation with

$$n^*/n = (m/m_1^*)(m_{\text{x.c.}}^*/m), \quad . \quad . \quad . \quad . \quad . \quad (5)$$

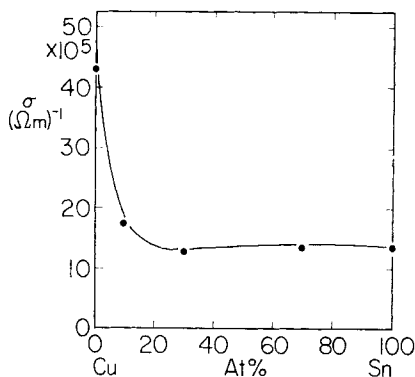
where (m_1^*/m) is the factor by which the density of states at the Fermi level is perturbed from its free-electron value by terms which contain the ionic pseudopotential *to first order*; $(m_{\text{x.c.}}^*/m)$ is the factor by which exchange and correlation would enhance the density of states even in an ideal jellium.

Fig. 10



Values of the effective valency, Z^* , for the copper-tin alloy system. The solid line represents the best fit to the data; the broken line represents free-electron behaviour.

Fig. 11



Values of $\sigma(0)$ for the copper-tin alloy system. The solid line shows the results for the d.c. conductivity measured by Roll and Motz (1957).

Corrections to the Drude formulae of order $(\hbar/2\tau E_F)^2$ or $(\hbar\omega/E_F)^2$ may certainly arise, but in the infra-red and visible they are expected to be rather trivial for the metals of interest here, with the possible exception of Hg. More significant, perhaps, are the various complicating factors associated with the surface which are liable to distort the apparent optical constants in a typical ellipsometric measurement. Among these may be mentioned (a) surface contamination, (b) the possibility that even when the surface is clear it is not ideally sharp, i.e. that $\sigma(\omega)$ does not change abruptly from its bulk value to zero, and (c) the possibility that electrons arriving at the surface are not specularly reflected but are scattered diffusely. Such factors could certainly distort the results by a few parts per cent in the visible region and might be blamed, therefore, for the fact that the

fitted values of $\sigma(0)$ (see table 1) are consistently less than $\sigma_{\text{d.c.}}$. As the frequency is decreased, however, and the depth to which the radiation penetrates the specimen increases, they should become progressively less important. In liquid Bi at 500°C, the penetration depth should increase from about 300 Å at a photon energy of 1 eV to about 1000 Å at 0.1 eV and one would expect the surface correction, if indeed these are significant, to diminish by a factor of roughly 3. Yet the curve for $\sigma(\omega)$ seems to remain horizontal over the whole of this range, consistently about 7% below $\sigma_{\text{d.c.}}$. No convincing explanation is available, therefore, for the low values of $\sigma(0)$.

Despite this difficulty the fitted values for n^*/n appear to be significant, since for the polyvalent metals they agree quite well with eqn. (5). The correlation is shown in table 2 where the experimental values for n^* for different temperatures have been averaged. The theoretical values listed for (m_1^*/m) are taken from Shaw and Smith (1969) and appear to be the best available, while the values for (m_{xc}^*/m) are from the work reviewed by Hedin and Lundqvist (1969). The result for In is that given by Hodgson (1962). It is worth noting that there does appear to be a small systematic decrease in n^* with increasing temperature (e.g. Ga, Sn, Pb) but no explanation for this behaviour has been found. Hodgson (1961) commented on a similar decrease for his results on Sn.

Values of n^*/n can be estimated for solid metals as well as liquid metals, by fitting the Drude formulae to results obtained in the infra-red at frequencies below the interband absorption peaks. It follows from the well-known absorption sum rule that if the interband absorption disappears on melting, then n^*/n must increase, and so indeed it does; the best result for n^*/n in solid Al for example, seems to be 0.67 (Ehrenreich, Philipp and Segall 1963) whereas it is about 0.83 in the liquid phase. The change on melting is of particular interest in the case of the noble metals Cu, Ag and Au. According to results quoted by Myers, Walldén and Karlsson (1968), the results available for the solid phase suggest that n^*/n is 0.69 ± 0.02 , 0.97 ± 0.05 and 0.95 ± 0.08 , respectively, so it increases on melting by about 10%, 8% and 6%. The increases expected from a simple theory discussed by Beaglehole (1966) and Miller (1969) amount to 8%, 4% and 7%. Within the large limits of error, experiment and theory are in agreement.

Table 2

Metal	n^*/n	m_1^*/m	$(m/m_1^*)(m_{\text{xc}}^*/m)$
Aluminium	0.82	1.03	0.95
Gallium	1.02	0.96	1.02
Indium	1.06	0.91	1.08
Tin	1.06	0.93	1.06
Lead	1.16	0.86	1.14
Bismuth	1.05	0.87	1.13

The results obtained for Cu-Sn alloys at 1150°C confirm the suggestion of Busch and Güntherodt (1967), based on Hall effect measurements at about the same temperature, that the NFE model is valid for this system. It is unfortunate that, because of difficulties with oxide contamination, no measurements were taken at lower temperatures ($\sim 550^\circ\text{C}$) where Enderby, Hasen and Simmons (1967) have found evidence from Hall effect measurements that some of the valence electrons may be attached to Sn ions in localized states, so that they do not contribute to the conduction process.

Little can be added to the controversy concerning the optical properties of liquid Hg. Bloch and Rice (1969) point out that all who have measured its reflectivity at normal incidence have obtained results in excellent agreement, over a very wide range of frequency, with free-electron Drude theory. They prefer to believe, therefore, that eqns. (1) with $n^* = n$ accurately represent the behaviour of Hg in the bulk, and that the discrepancies revealed by the ellipsometric measurements are to be blamed upon peculiarities of the surface of Hg. They have suggested an interesting model for the surface which could be consistent with all the results hitherto obtained, whether from reflectivity measurement or by ellipsometry. There is a good chance that by minor adjustment of the several parameters involved it could be rendered equally consistent with the new data presented in fig. 9.

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